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Laboratory – (Advanced Java Programming)**AIM**

Design a Swing form using JTextField and JTextArea to accept and display user input.

Theory**1. INTRODUCTION**

Java is a versatile, object-oriented programming language that is widely used for developing various types of applications such as console-based applications, web applications, mobile applications, enterprise systems, and desktop applications. Desktop applications require a graphical user interface (GUI) that allows users to interact with the system using visual elements like windows, buttons, text boxes, checkboxes, and tables. To support GUI development, Java provides libraries such as AWT and Swing.

Java Swing is a part of the Java Foundation Classes (JFC) and is used to create platform-independent, window-based applications. Swing components are lightweight and written completely in Java, which ensures consistent behavior across different operating systems. Swing follows an event-driven programming model, where user actions generate events that are handled by event listeners.

Swing provides a rich set of components such as JTextField, JTextArea, JCheckBox, JPanel, JTable, JButton, and JLabel, which help developers create interactive and user-friendly interfaces. These components can be arranged using layout managers and grouped logically to design structured applications.

In this practical, multiple Swing applications are designed to understand user input handling, multiple option selection, component organization using panels, event handling using ActionListener, and displaying data in tabular form using JTable. These concepts are fundamental for developing real-world Java desktop applications.

2.1 Swing Form Using JTextField and JTextArea to Accept and Display User Input

JTextField and JTextArea are commonly used Swing components for handling textual input and output. JTextField allows the user to enter a single line of text, while JTextArea allows multiple lines of text. These components are essential in form-based applications such as login pages, registration forms, and feedback systems.

A JLabel is usually used along with JTextField to describe the purpose of the input field. The text entered by the user can be retrieved using the getText() method and displayed in a JTextArea using the setText() method. JTextArea is often placed inside a JScrollPane to allow scrolling when the text content is large.

This concept helps in understanding how user input is accepted, processed, and displayed dynamically in a GUI application. It also introduces interaction between multiple Swing components through event handling.

2.2 Swing Application Using JCheckBox to Select Multiple Options

JCheckBox is a Swing component that allows users to select one or more options from a given list. Unlike radio buttons, checkboxes do not restrict the user to a single choice. Each JCheckBox operates independently and can be selected or deselected.

JCheckBox is commonly used in applications where multiple selections are required, such as selecting hobbies, skills, preferences, or features. The state of a checkbox can be checked using methods like isSelected().

When the user selects multiple checkboxes and performs an action such as clicking a button, the application can gather all selected options and display them. This demonstrates conditional logic and user-driven input handling in Swing applications.

2.3 Designing a Swing GUI Using JPanel to Organize Components

JPanel is a lightweight container used to group and organize Swing components logically. It helps in structuring complex user interfaces by dividing them into smaller sections. Each JPanel can have its own layout manager and components.

Using multiple panels improves the readability, maintainability, and appearance of the GUI. For example, one panel may contain input fields, another may contain buttons, and another may display output. Panels can be nested inside other panels to create structured layouts.

JPanel plays a crucial role in building scalable and professional-looking Swing applications. It allows developers to manage components efficiently and maintain consistent alignment and spacing.

2.4 Event Handling in Swing Using ActionListener

Swing applications follow an event-driven programming model. An event is generated when the user interacts with the GUI, such as clicking a button or pressing the Enter key. ActionListener is one of the most commonly used listener interfaces in Swing.

Components like JButton and JTextField generate(ActionEvent) events. By implementing ActionListener and overriding the actionPerformed() method, the programmer can define how the application responds to user actions.

Event handling allows the application to be interactive and responsive. It connects the user interface with the application logic, enabling real-time processing of user input. Understanding ActionListener is essential for building functional Swing applications.

2.5 Swing Application Using JTable to Display Tabular Data

JTable is a powerful Swing component used to display data in rows and columns. It is commonly used in applications that require structured data representation, such as student records, employee details, billing systems, and inventory management systems.

JTable works with a data model, usually implemented using DefaultTableModel. The model stores the data, while JTable is responsible for displaying it. JTable supports features like scrolling, column resizing, and data selection.

To display large datasets, JTable is placed inside a JScrollPane. JTable helps in presenting data clearly and efficiently, making it an important component in real-world desktop applications.

3. Conclusion:

4. Questions:

1. What is Java Swing and why is it used for GUI development?
2. What is the difference between JTextField and JTextArea?
3. How is user input accepted from JTextField and displayed in JTextArea?
4. What is the role of ActionListener in a Swing application?